

Hierarchical Models for RNA-seq

Michael Love

Hierarchical Models for RNA-seq

Michael Love

Dept of Genetics

Dept of Biostatistics

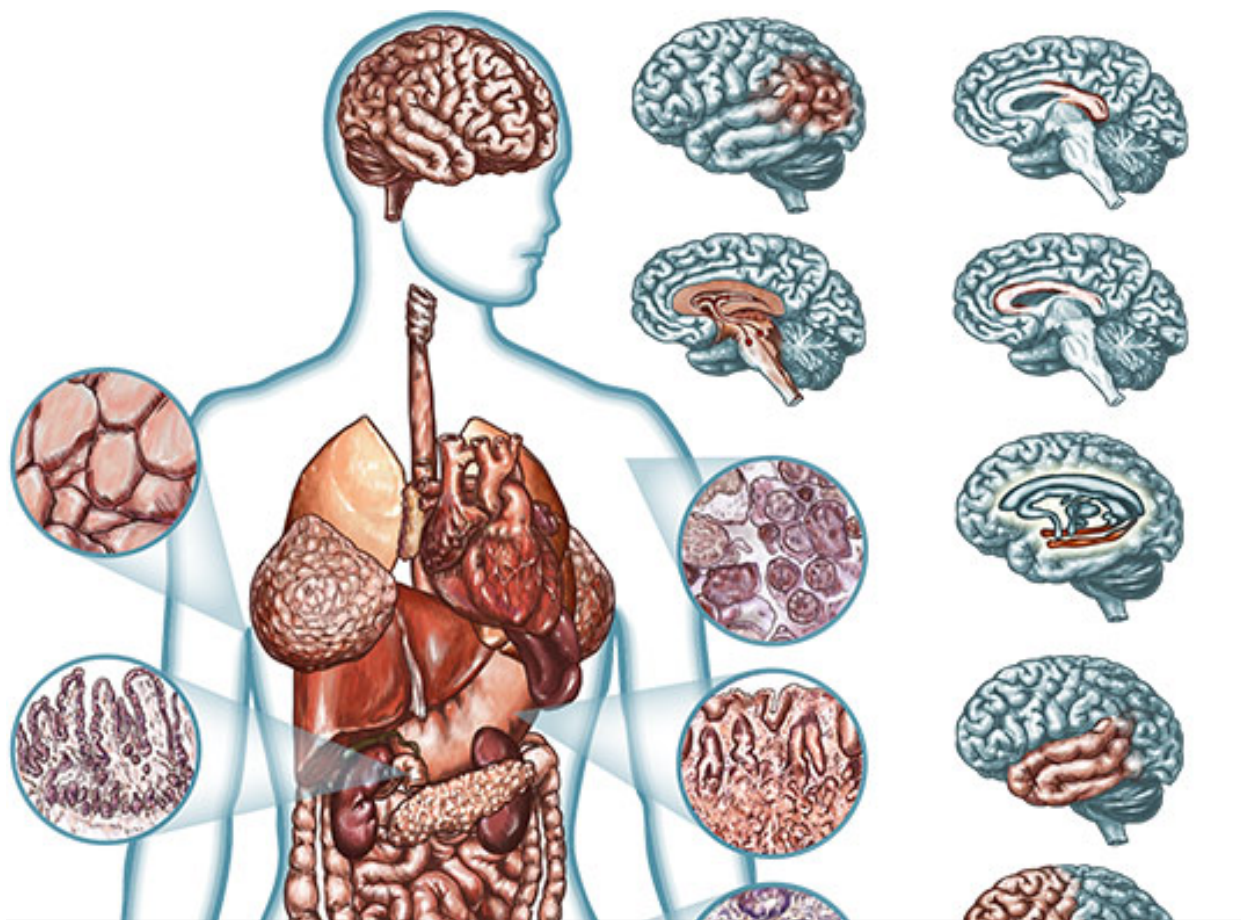
Outline

- Why RNA
- Motivation of hier. model for continuous gene expression
- Extension to sequence count data
- Questions

Why measure RNA as phenotype

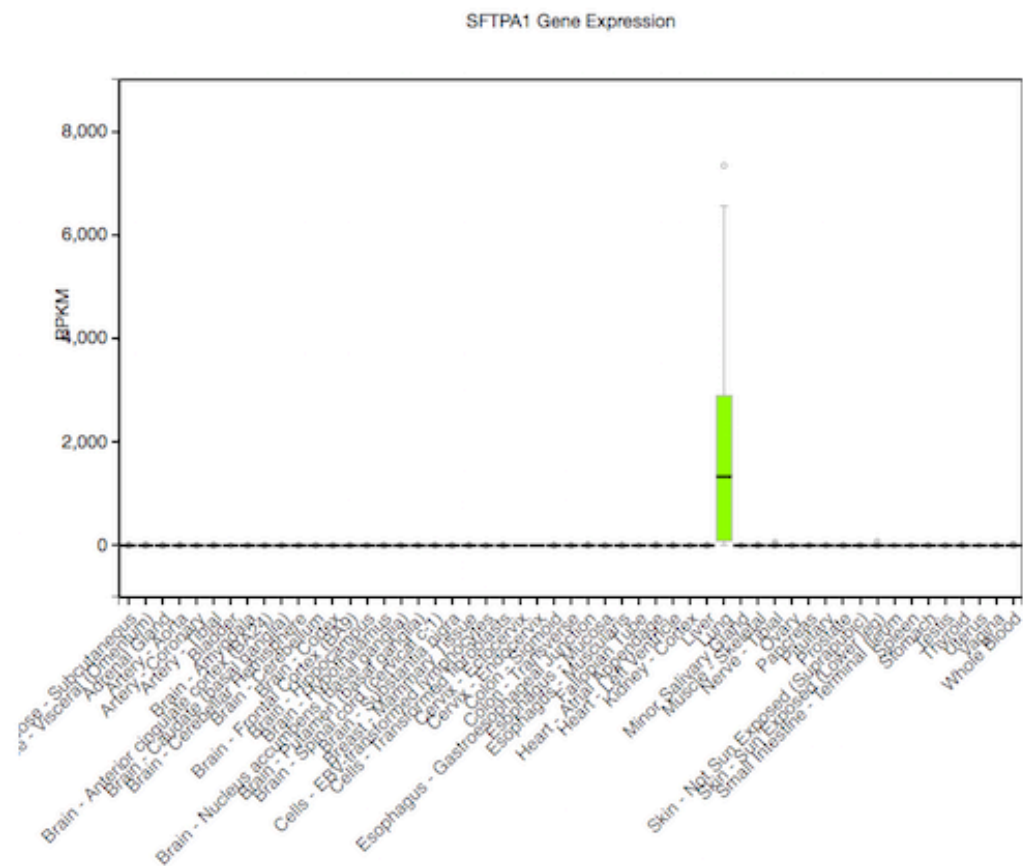


Why measure RNA: tissue diversity



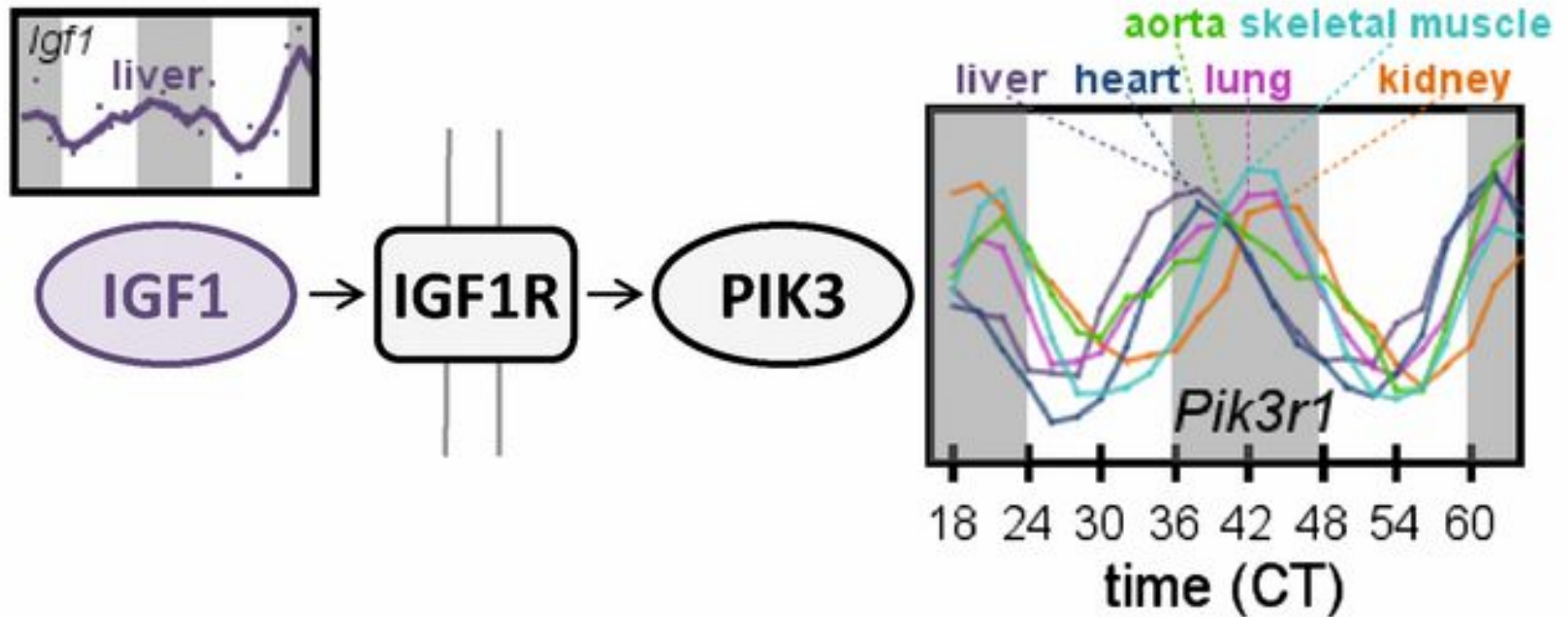
Roadmap Epigenomics

Why measure RNA: tissue diversity



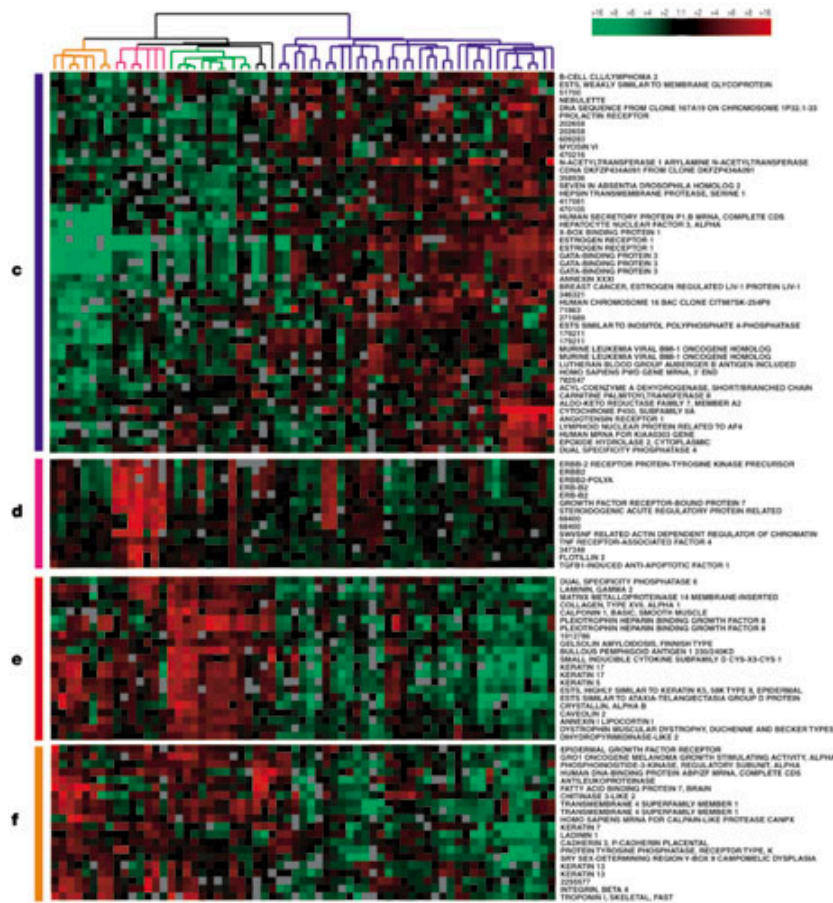
GTEx

Why measure RNA: within tissue over time



Zhang, et al. Circadian gene expression atlas (2014)

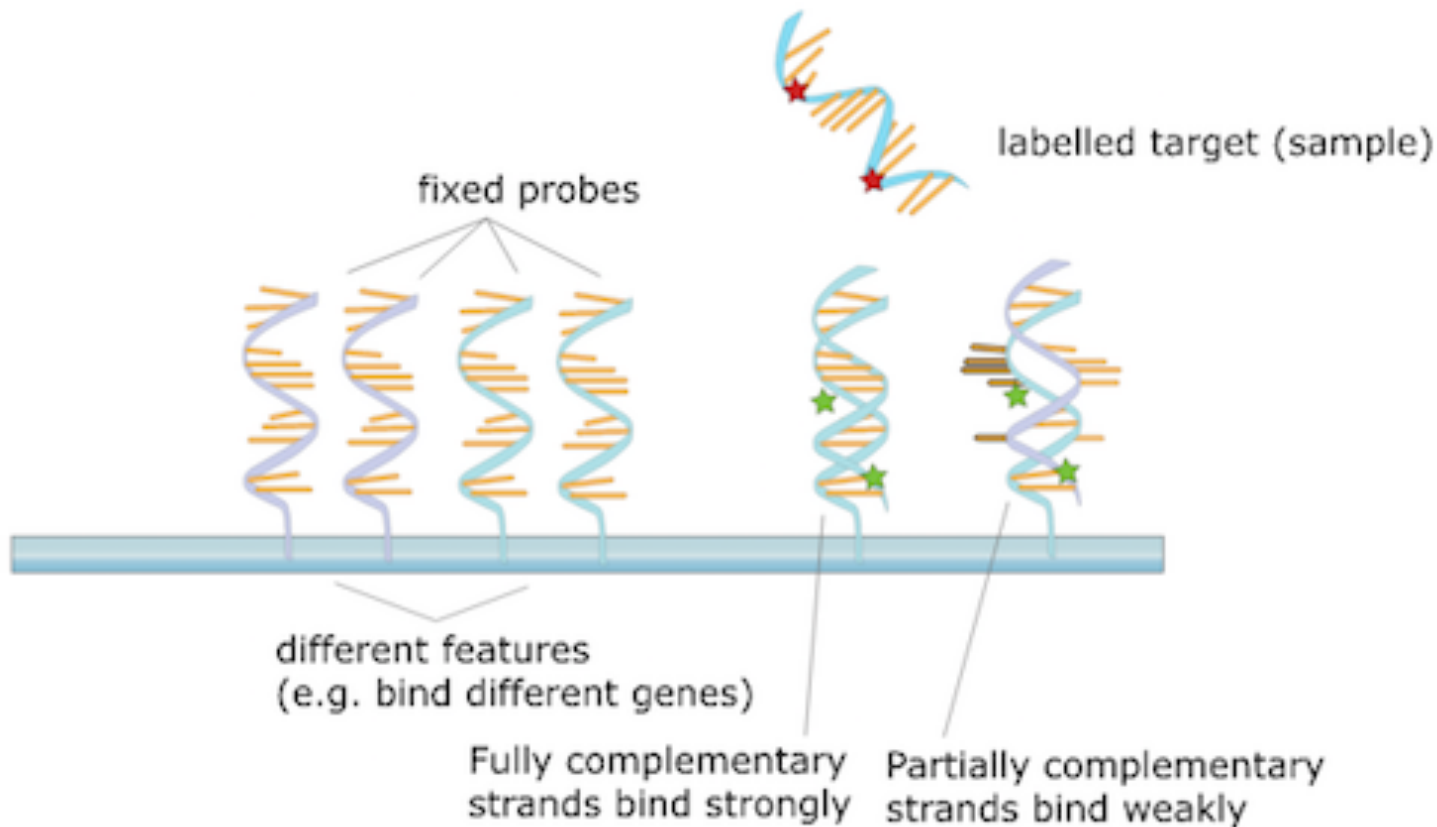
Why measure RNA: disease sub-types



Perou, et al. Molecular portraits of human breast tumours (2000)

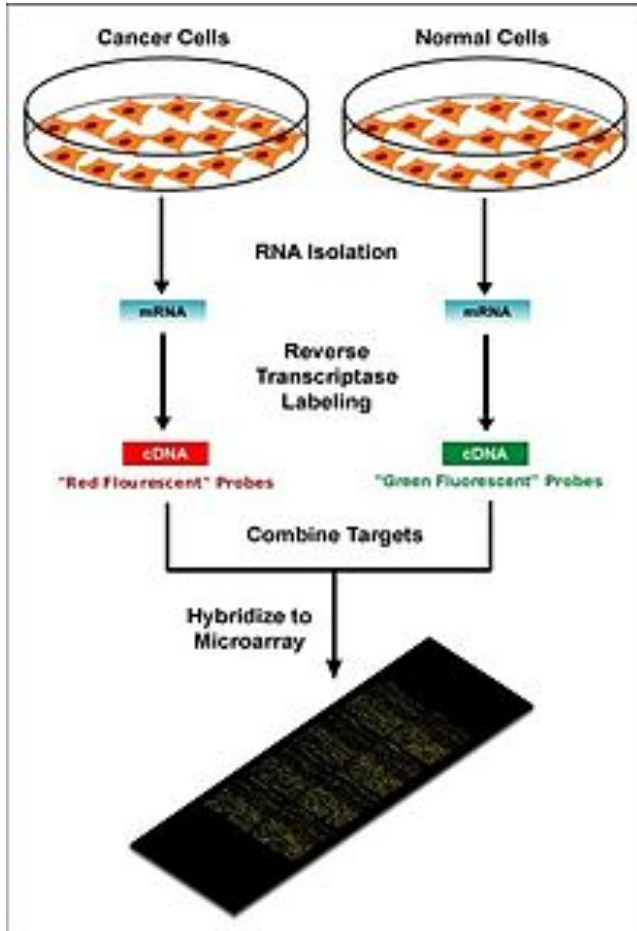
Step back: pre-sequencing

Before sequencing was microarray



Step back: pre-sequencing

Signal was captured light (positive, “continuous”)



Formalize

- Formalize as a simple statistical model
- Observed data X_{ij} as a R.V. following a distribution
- Distributional parameters of interest

Motivating problem

- Gene expression for $i=1,\dots,N$ genes and $j=1,\dots,M$ samples
- log of “true” expression values (non-negative, long tail)
- Suppose 2 equal sized groups of samples: “A” and “B”

$$X_{ij} \sim N(\mu_{ij}, \sigma_i)$$

$$\mu_{ij} = \mu_{i0}, \quad j \in A$$

$$\mu_{ij} = \mu_{i0} + \delta_i, \quad j \in B$$

$\delta_i \neq 0$ implies DE (differential expression)

Note σ_i

Different genes i have different amount of variability.

$$X_{ij} \sim N(\mu_{ij}, \sigma_i)$$

$$\mu_{ij} = \mu_{i0}, \quad j \in A$$

$$\mu_{ij} = \mu_{i0} + \delta_i, \quad j \in B$$

Goal of differential expression

- Find a set of genes for which $\delta_i \neq 0$
- We want to target a false discovery rate (FDR) cutoff
- For genes in our set G at FDR threshold z

$$E \left(\sum_{i \in G} 1_{\{\delta_i=0\}} \right) \leq |G| z$$

Is this realistic?

- Can we accomplish this if all $\delta_i \neq 0$
 - no, because methods often rely on global scaling normalization
- Are any $\delta_i = 0$?
 - maybe not, but many are very small for controlled experiment

Is this realistic?

- What about σ_i for both groups?
 - often this is enough, larger variance dominates
 - not for single cell experiments
- More complex parametric models: [baySeq](#)
- Non-parametric: [SAM](#) / [SAMseq](#)

Back to the model

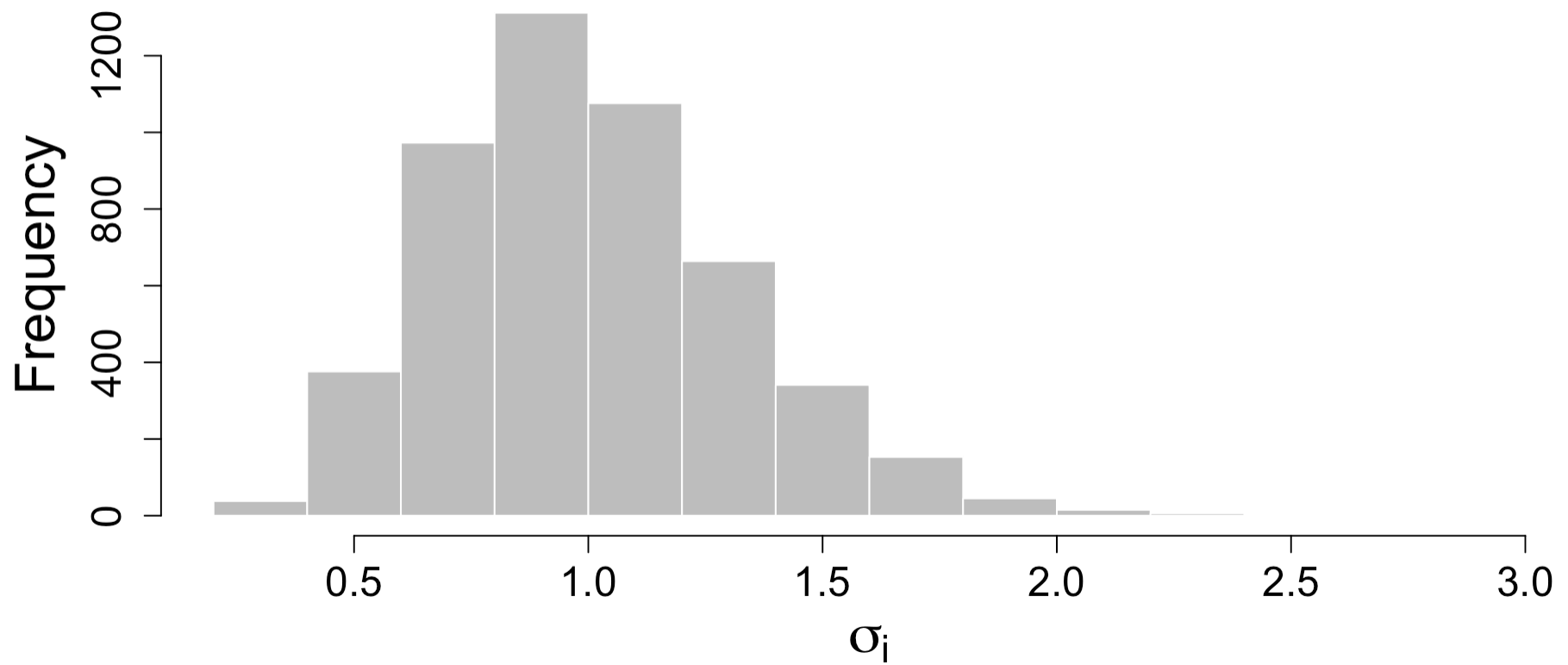
$$X_{ij} \sim N(\mu_{ij}, \sigma_i)$$

$$\mu_{ij} = \mu_{i0}, \quad j \in A$$

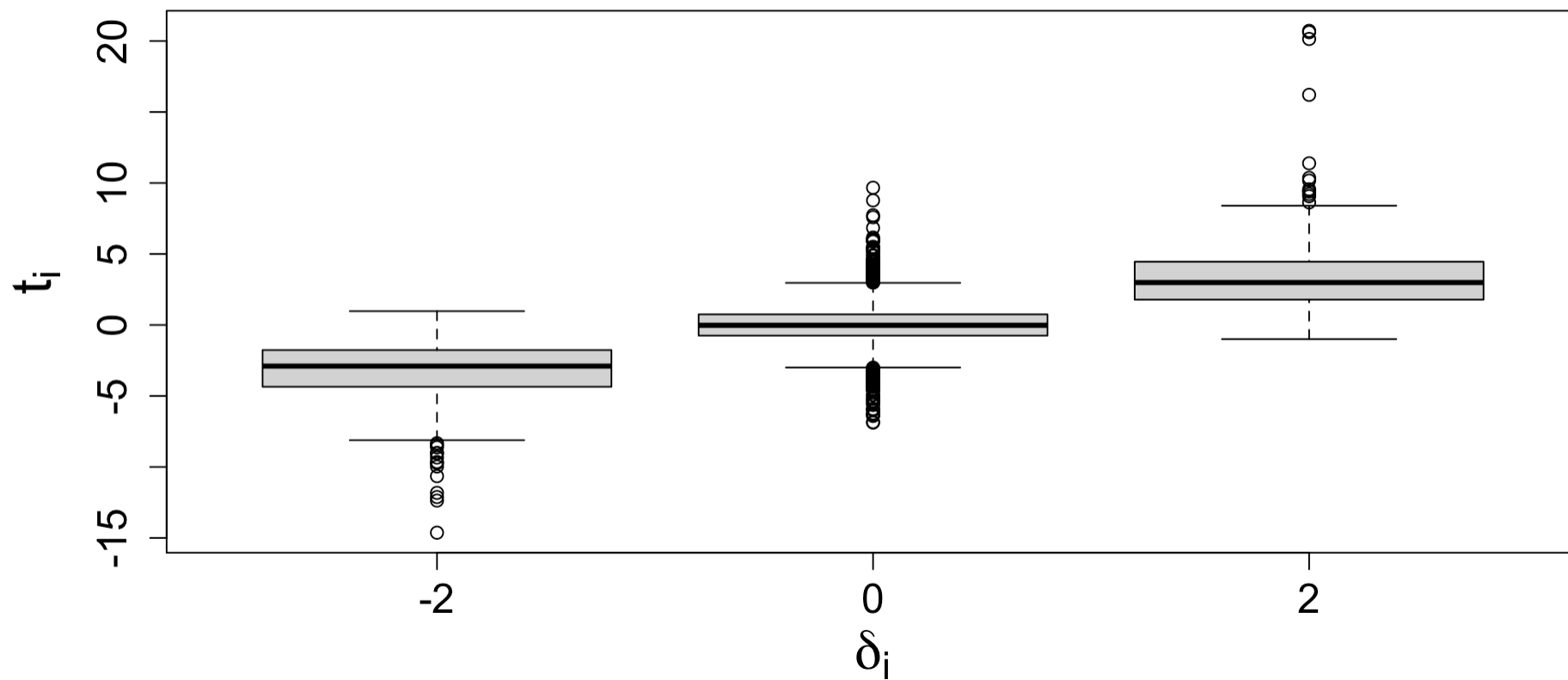
$$\mu_{ij} = \mu_{i0} + \delta_i, \quad j \in B$$

- $N = 5000, M = 6$
- $\delta_i = 0$ for 90%
- $\delta_i = \pm 2$ for 10%
- $\sigma_i \sim \Gamma(10, 10)$

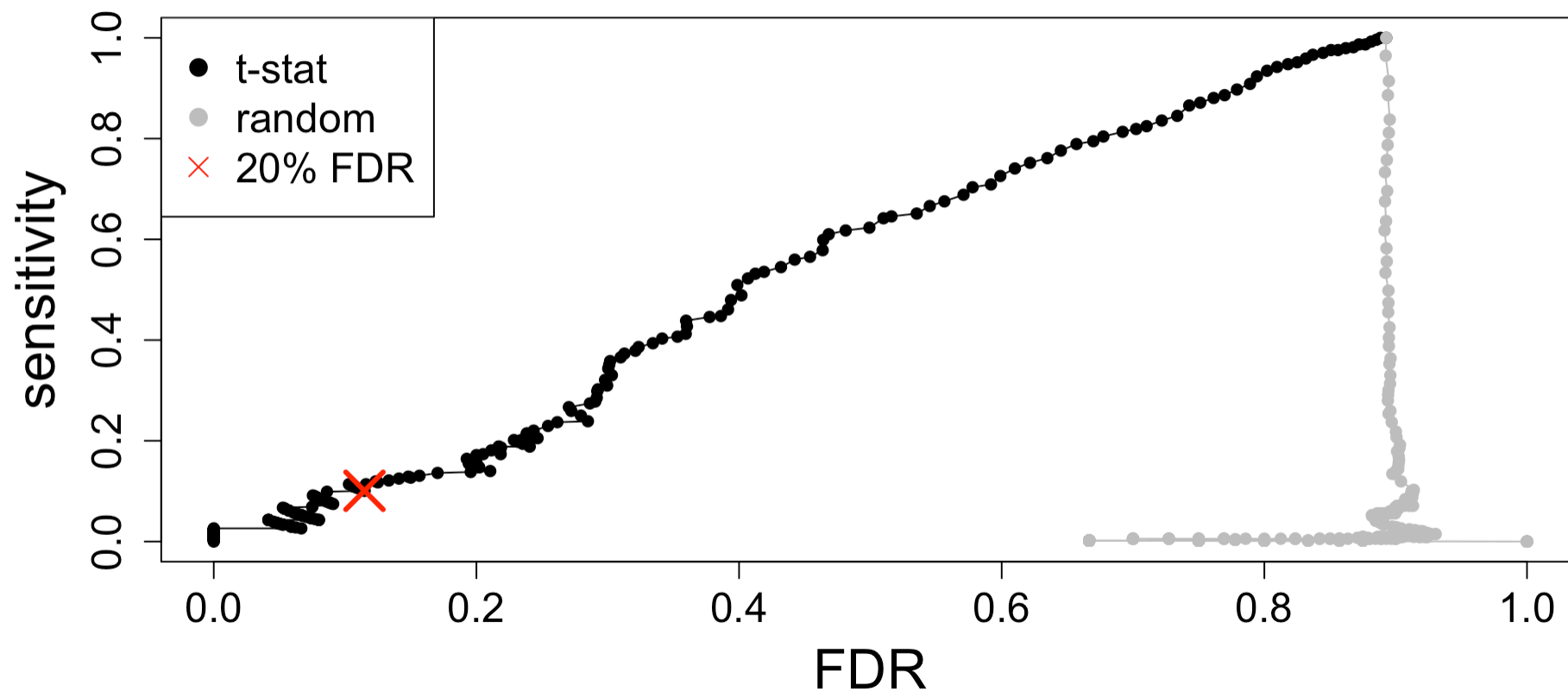
Distribution of σ_i



Try simple row t-tests

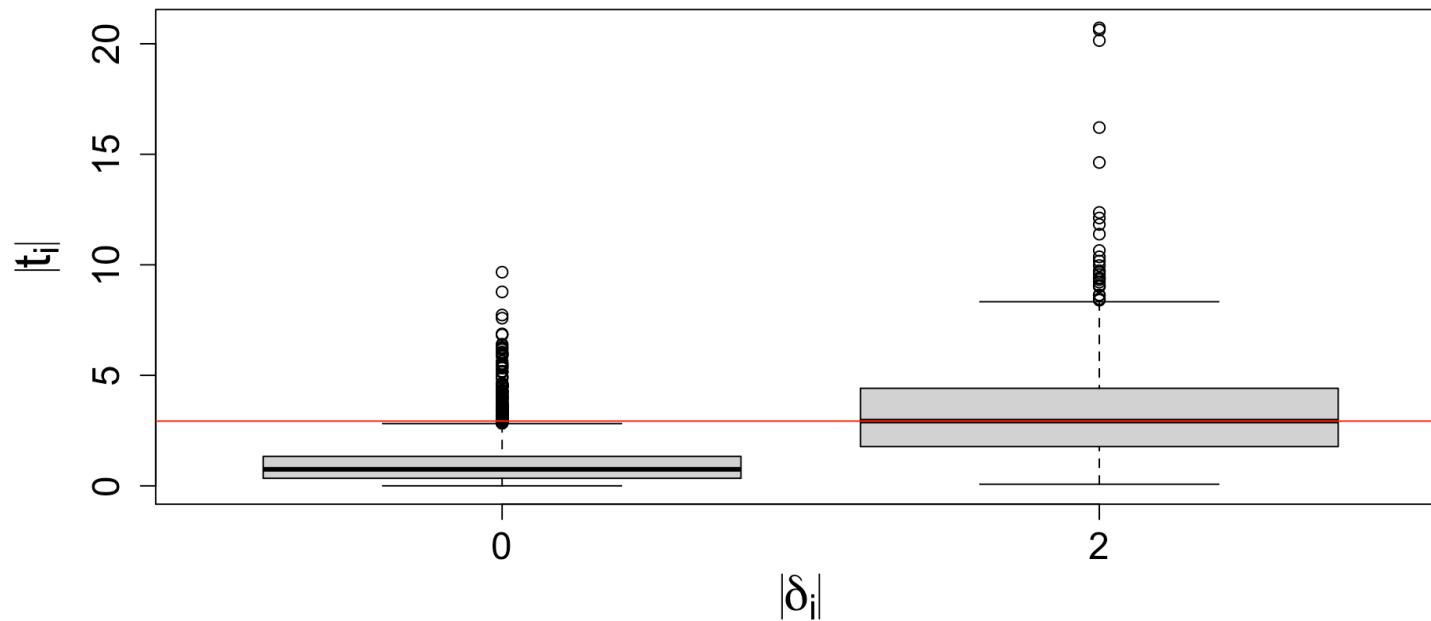


Just looking at ranks



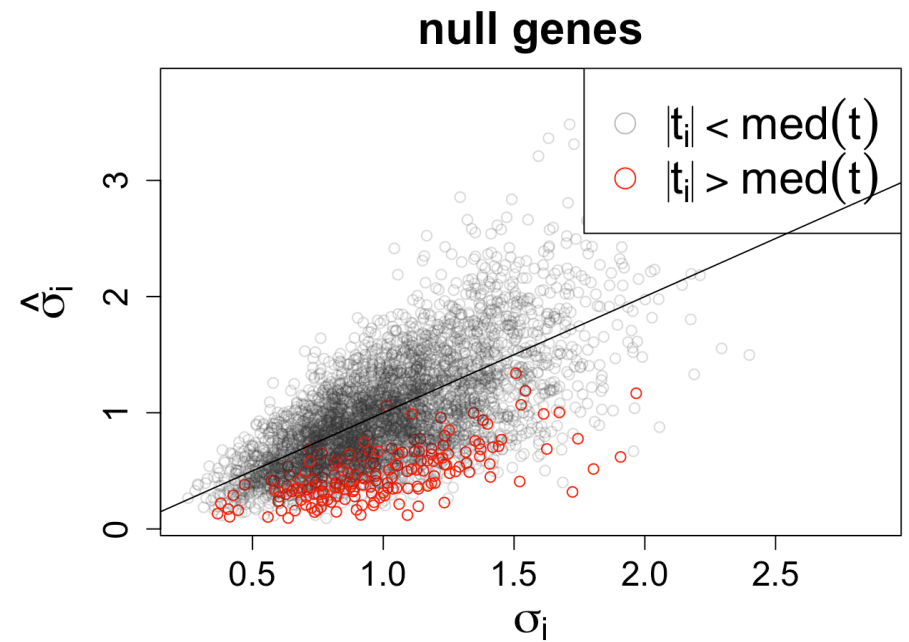
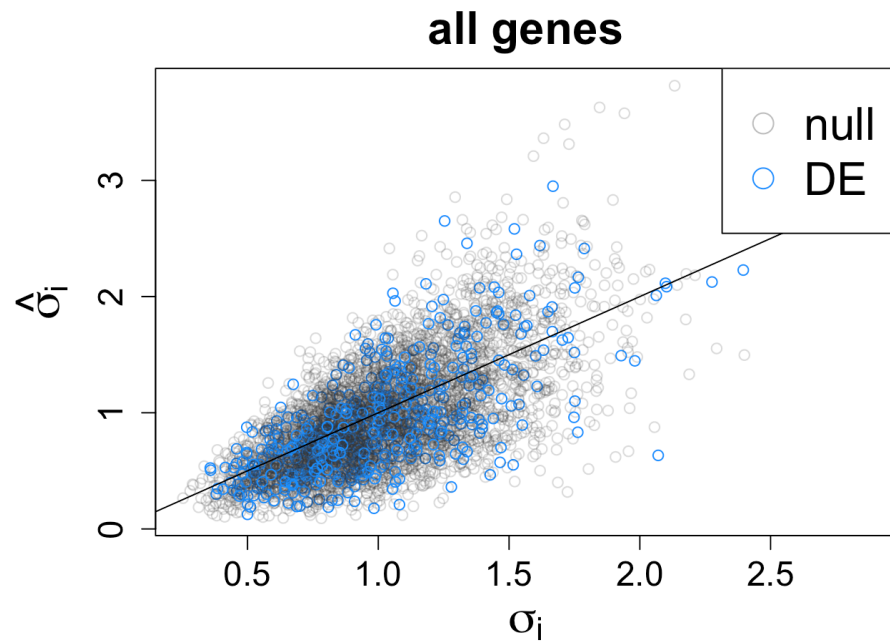
Characterize the false positives

$$\text{med}(t) \equiv \text{median}(|t_i|) \text{ for } i : \delta_i \neq 0$$



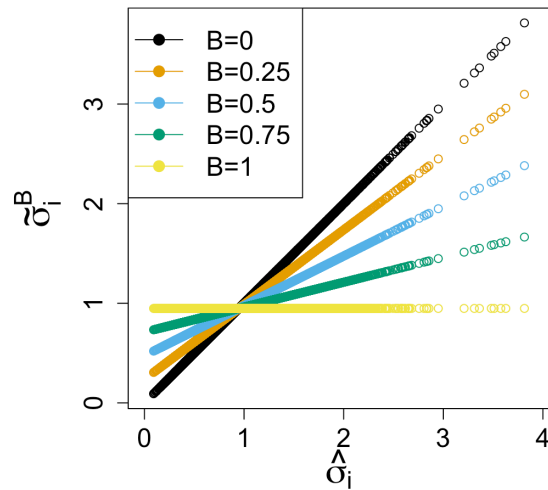
Estimates of σ_i

$$\text{med}(t) \equiv \text{median}(|t_i|) \text{ for } i : \delta_i \neq 0$$

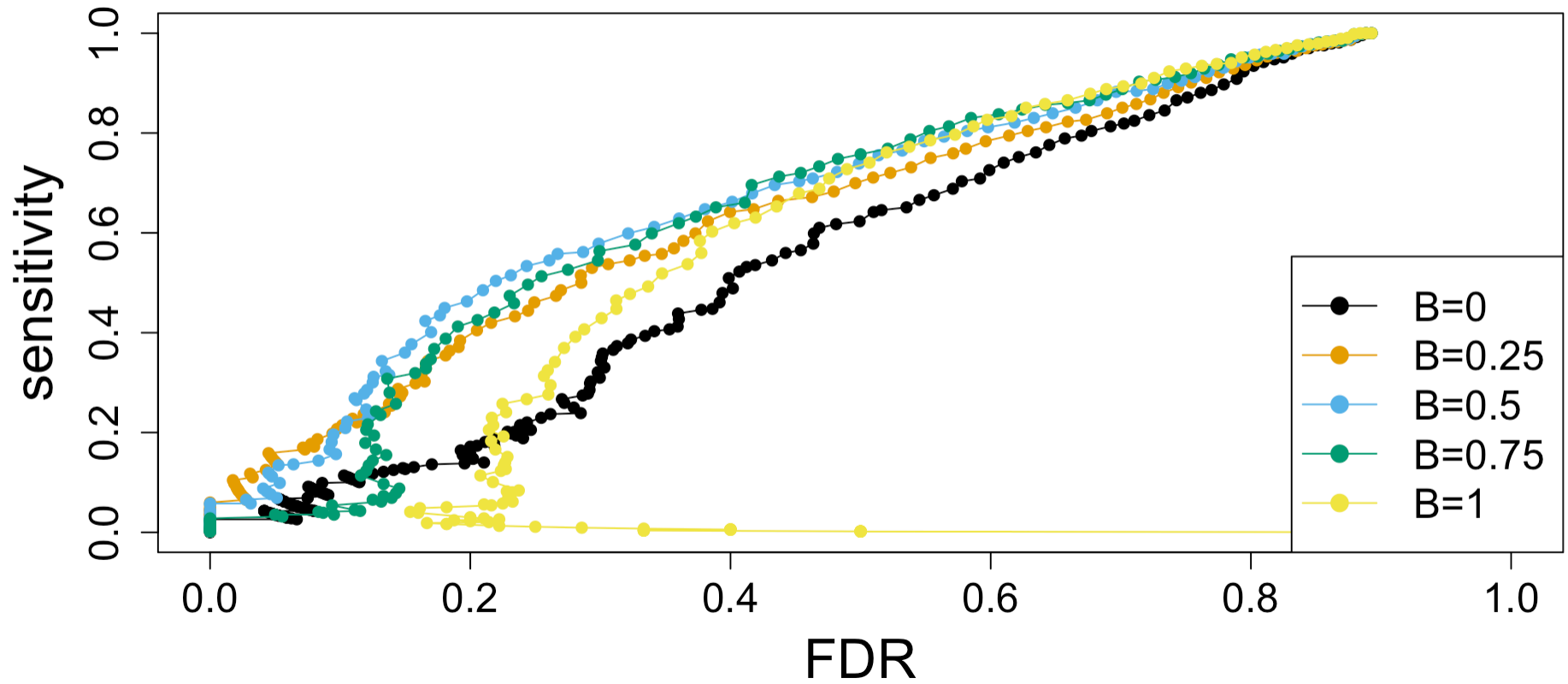


New estimator for σ_i

$$\bar{\sigma} = \frac{1}{N} \sum_{i=1}^N \hat{\sigma}_i$$
$$\tilde{\sigma}_i^B = B\bar{\sigma} + (1 - B)\hat{\sigma}_i$$

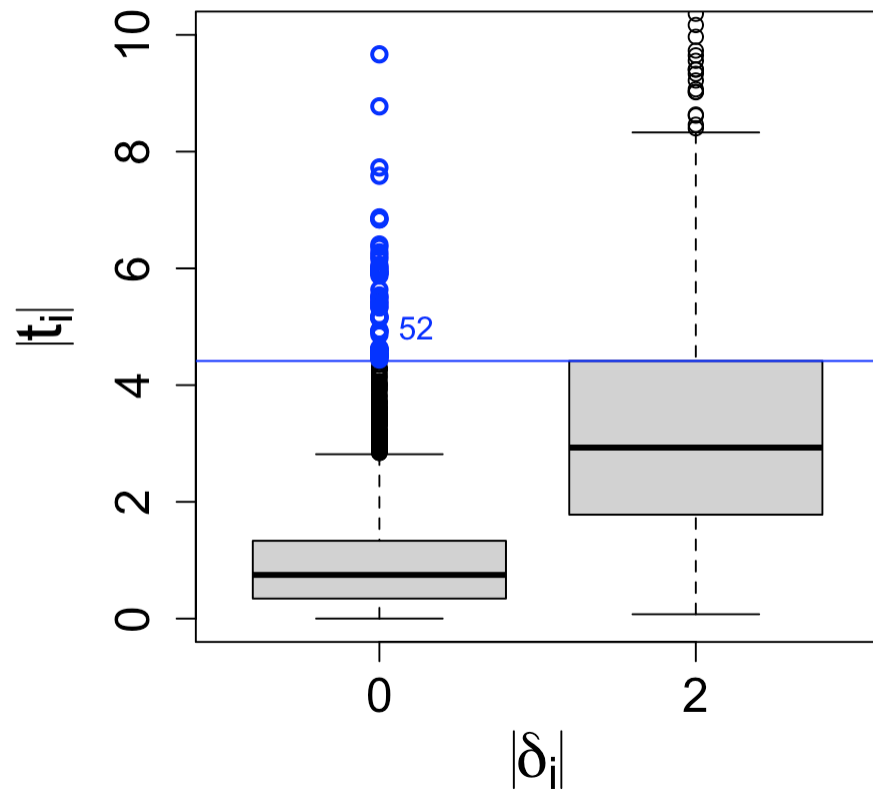


New estimator performance by rank

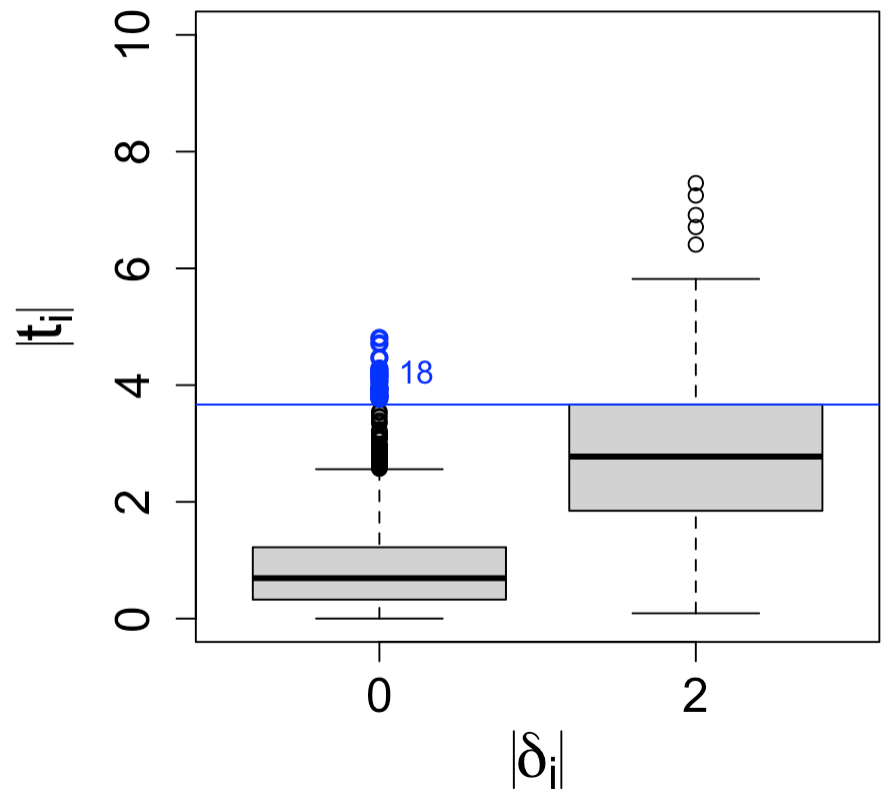


New estimators by $|t|$

orig



B=1/2

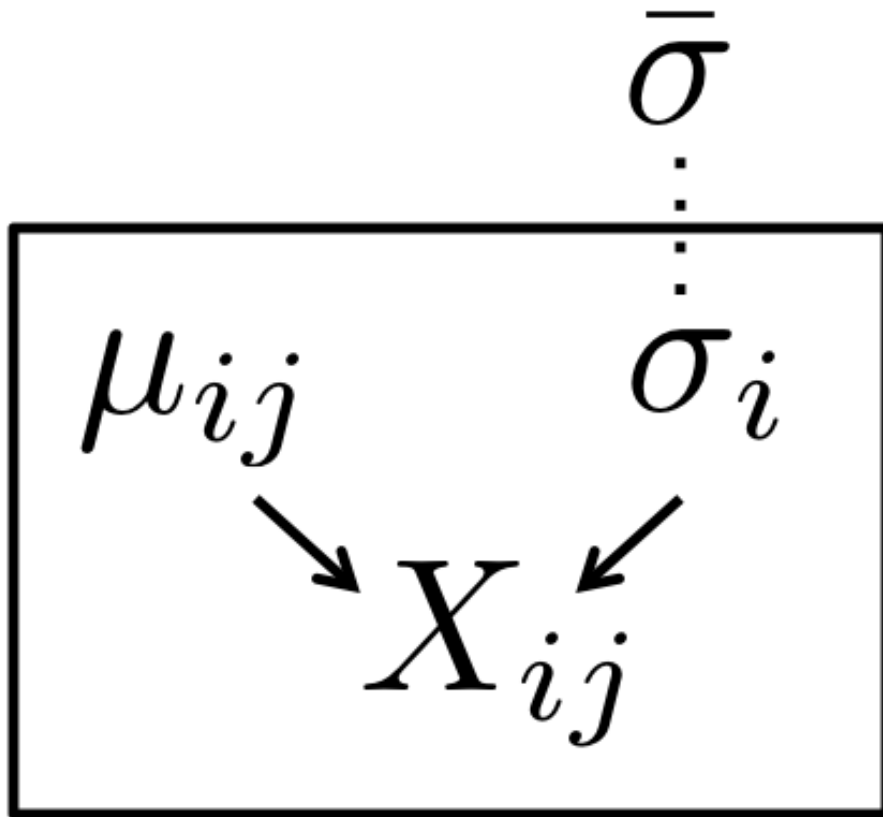


Summary

- Top false positives were coming from genes with too low $\hat{\sigma}_i$
- Replace $\hat{\sigma}_i$ with an estimate which is closer to $\bar{\sigma}$
- Depending on “close”, new estimator dominates at all thresholds

How is this hierarchical?

Not your standard diagram, need to formalize



limma

- [Smyth, G. K. \(2004\)](#) Linear models and empirical Bayes methods for assessing differential expression in microarray experiments
- Developed the hierarchical model introduced by Lonnstedt and Speed (2002) for single sample into method for any experiment represented as linear model

$$\frac{1}{\sigma_i^2} \sim \frac{1}{d_0 \sigma_0^2} \chi_{d_0}^2$$

Why inverse χ^2 ?

- Conjugacy provides closed form solution
- Posterior mean for $1/\sigma_i^2$ given $\hat{\sigma}_i^2$ is $1/\tilde{\sigma}_i^2$ with

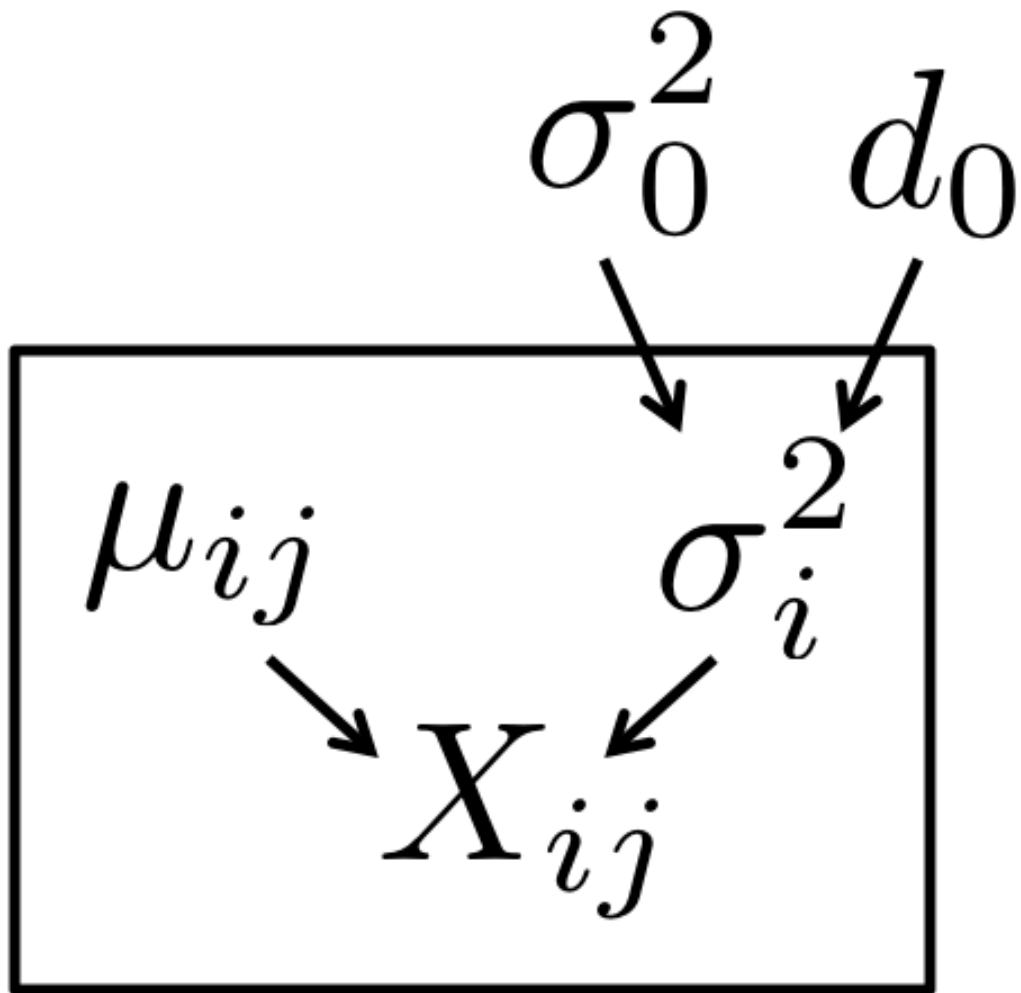
$$\tilde{\sigma}_i^2 = \frac{d_0 \hat{\sigma}_0^2 + d_i \hat{\sigma}_i^2}{d_0 + d_i}$$

And d_i as the standard residual degrees of freedom

Note that d_0 controls B

$$\begin{aligned}\tilde{\sigma}_i^2 &= \frac{d_0 \hat{\sigma}_0^2 + d_i \hat{\sigma}_i^2}{d_0 + d_i} \\ &= \left(\frac{d_0}{d_0 + d_i} \right) \hat{\sigma}_0^2 + \left(\frac{d_i}{d_0 + d_i} \right) \hat{\sigma}_i^2 \\ &= B \hat{\sigma}_0^2 + (1 - B) \hat{\sigma}_i^2\end{aligned}$$

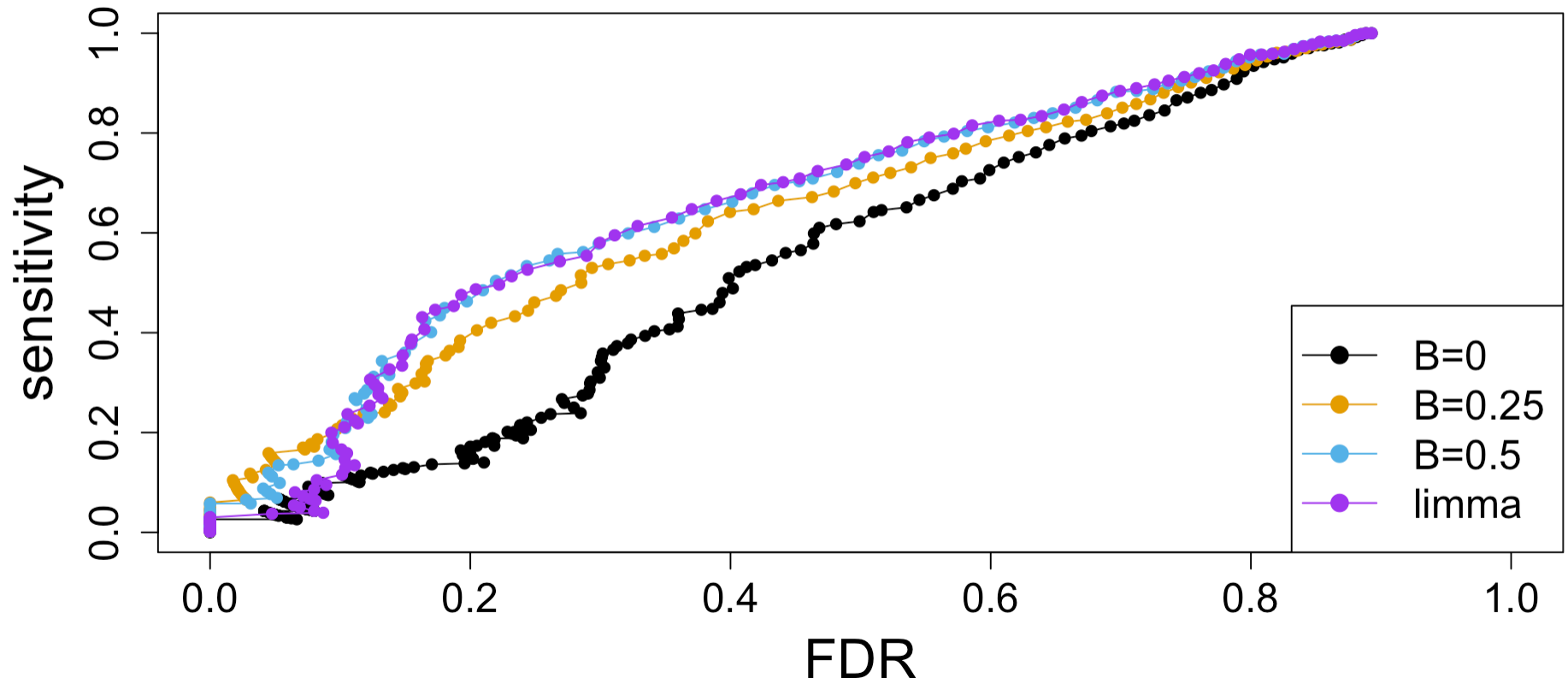
Proper hierarchical model



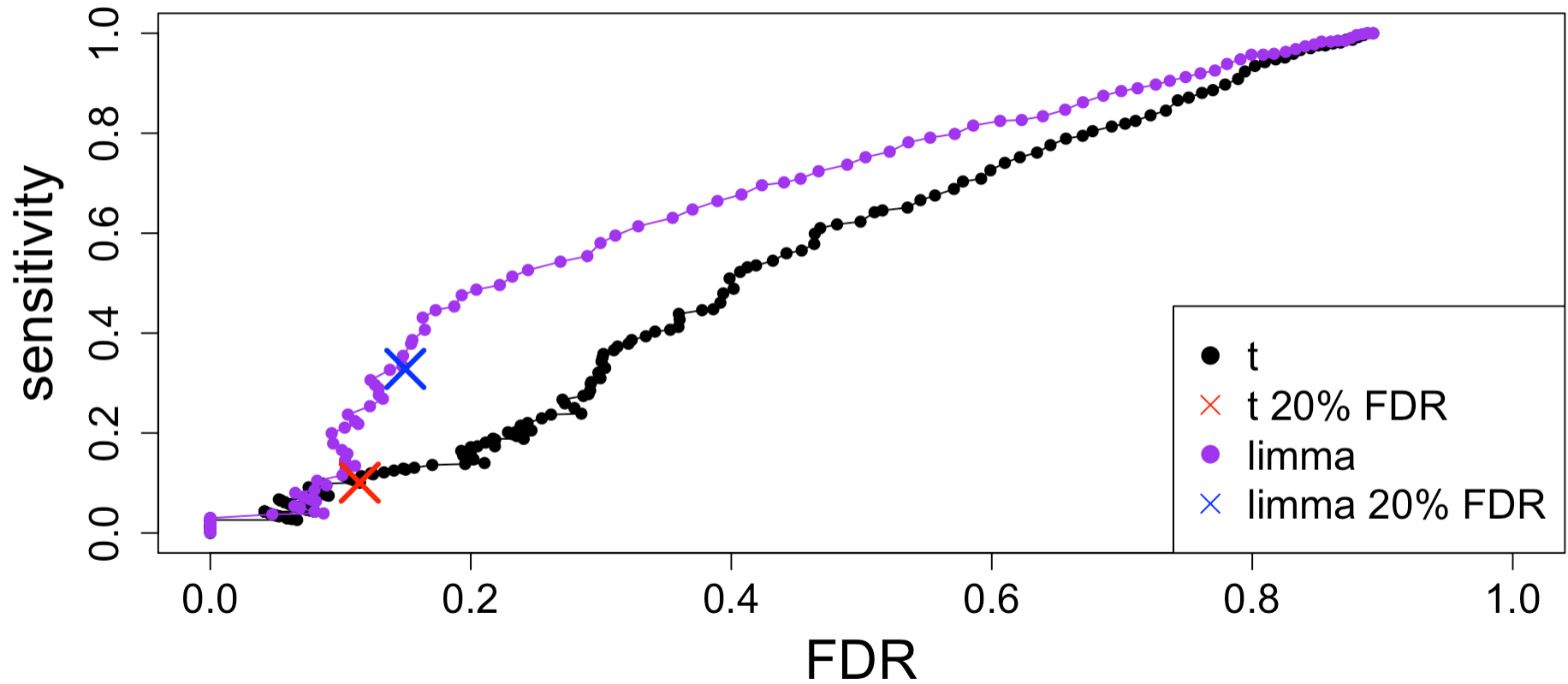
Estimation of hyperparameters

- Need to estimate $d_0, \hat{\sigma}_0^2$, which control strength and location of *shrinkage* or *moderation*
- $d_0, \hat{\sigma}_0^2$ estimated via first two moments of $\log \hat{\sigma}_i^2$
- (Also need to estimate ν_0 , another parameter giving variance of coefficients)

limma vs. naive estimators by rank



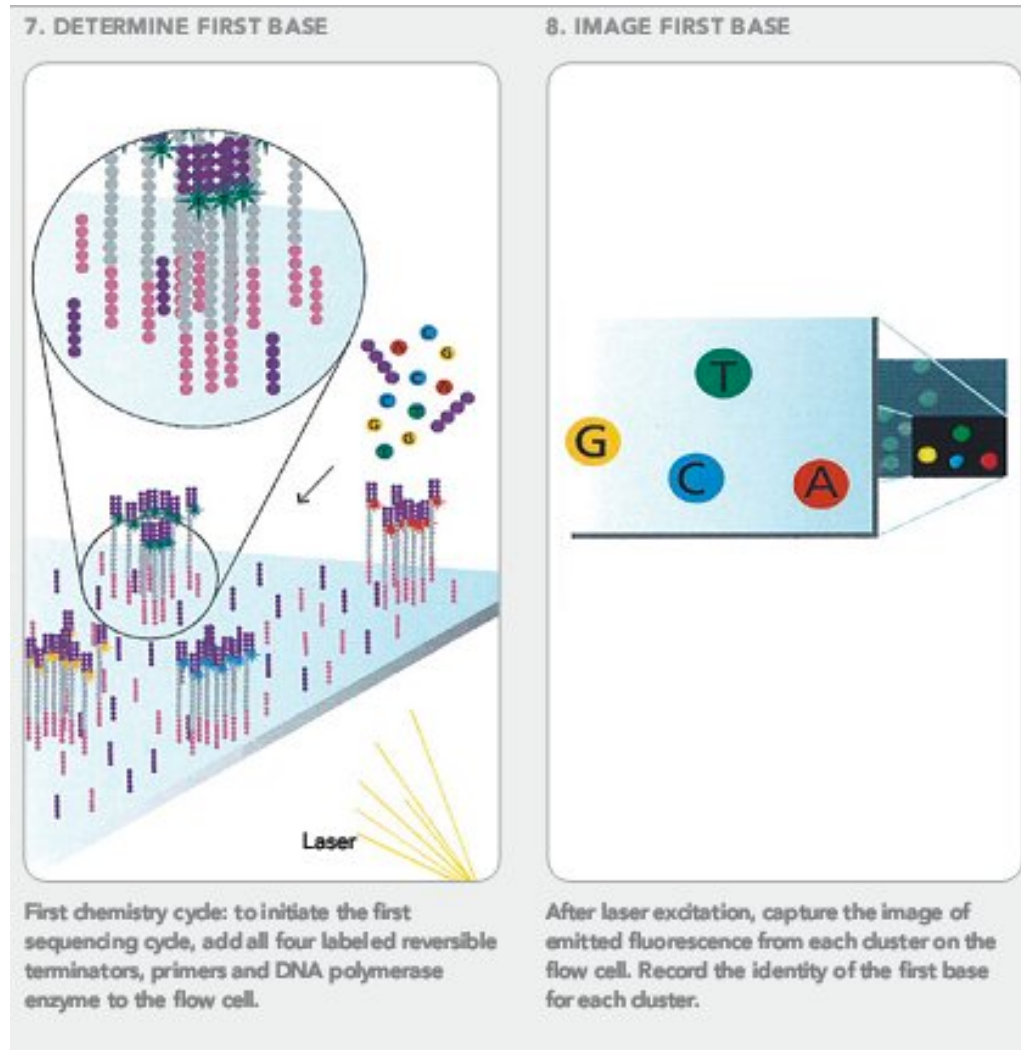
Operating characteristics (FP & FN)



Summary

- limma provides a hierarchical model for moderation of variance estimates in the context of linear models
- Avoids false positives from under-estimation of variance
- Also addresses the gain in degrees of freedom from moderation

RNA-seq: counting molecules



RNA-seq: counting molecules

@SRR1265495.1 1/1

CTTTGCCCCGCGTGTCAGACTCCATCCCTCCTCTGCCGCCACCGCAGCAGCCACAGGCAGAGGAGGACGAGGACGACTGG
+

CCCFHHFHHHHIJJIIJJIIJJJJJJJGIIJJIIJJJJJJJEFHDEFFFECDDBDD?BB?

B@BDDD@;9<BBDBCB@A2<?BDDDBDBD@CDDC

@SRR1265495.2 2/1

CCTGGCTGTGTCCATGTCAGAGCAATGGCCCAAGTCTGGGTCTGGGGGGGAAGGTGTCATGGAGCCCCCTACGATTCCC
+

@C@FFFDFFFHHHGGIIAGHI9GIIIIIGIIIIIGI@@FHGDDDH@GGIBB05?B>ACCCCCCB<?

BBCCBBBBCBCDC3>AC<BB<?CBBB?<@CCCA:@

@SRR1265495.3 3/1

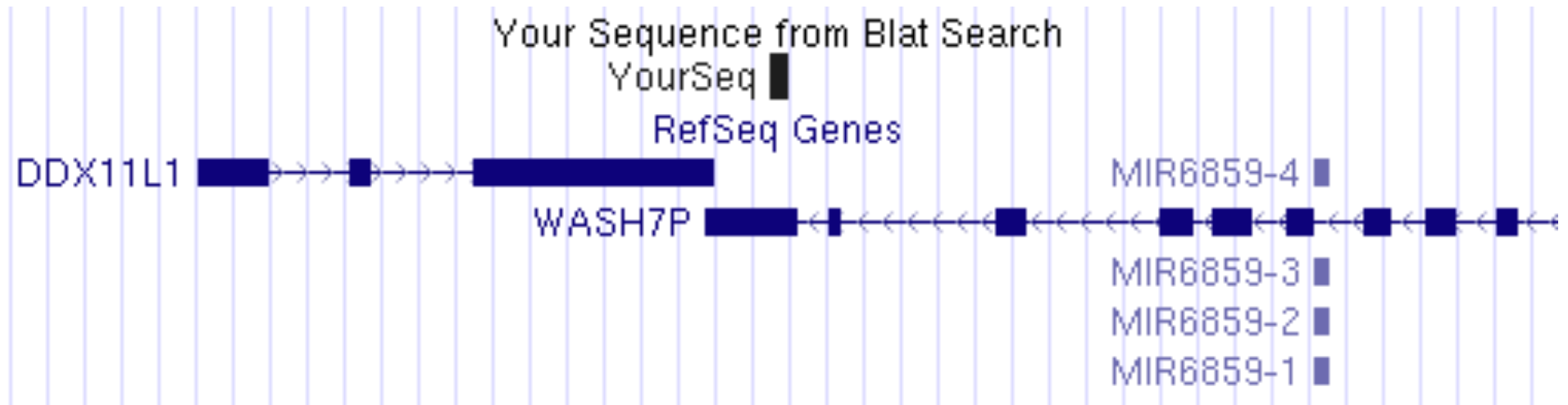
CTGTGTCCATGTCAGAGCAATGGCCCAAGTCTGGGTCTGGGGGGGAAGGTGTCATGGAGCCCCCTACGATTCCCAGTCG
+

@C@DFFFFGGHDHIIEGDHCGGHIJGEHIIJIIJIEEHGIGEGDDB@@@CDDDEDDDDDBDDDDDDDDDBCCDDCCDB
@DDDDCDDCC@A~

for ~30 million reads (often pairs of reads)

RNA-seq: counting molecules

Align to genome or transcriptome



RNA-seq: counting molecules

- Now for each gene and each sample we can obtain a *count* or *estimated count* of fragments
- Why estimated? Because some fragments cannot be uniquely associated with genes or isoforms
- Fast algorithms for probabilistically assignment:
 - [kallisto](#) (2016)
 - [salmon](#) (2017)
- Assume we have integer counts K_{ij} of unique fragments or from rounding estimated counts

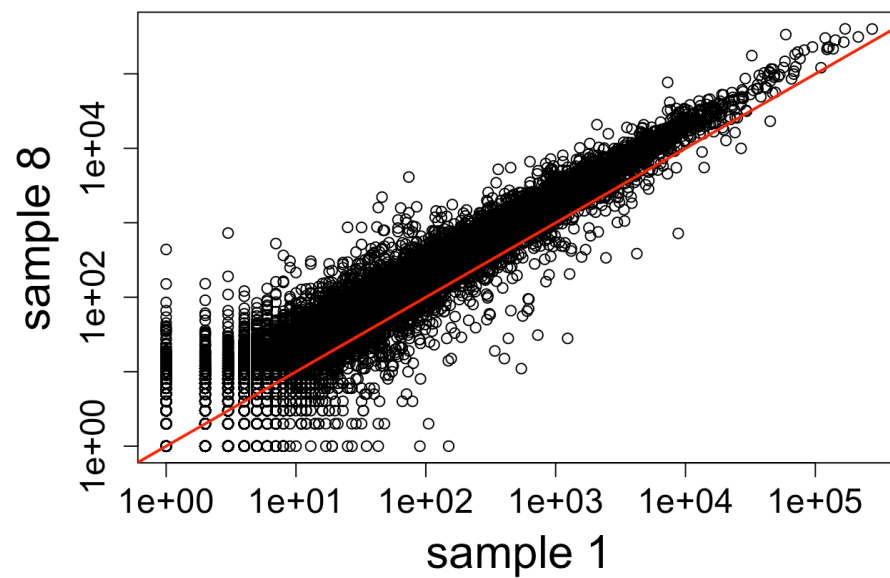
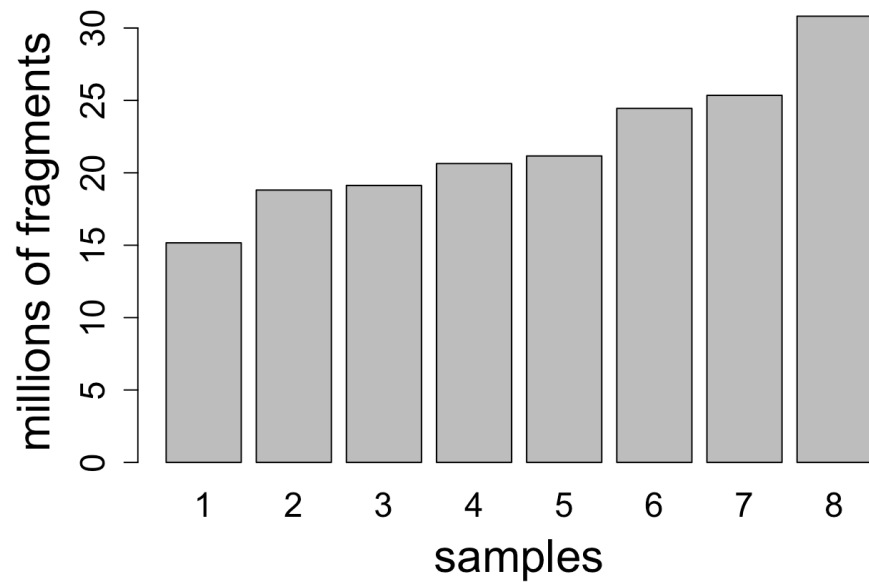
Counts

1. Either model data with count distributions and inference with GLM
 - [DESeq2](#) (2014)
 - [edgeR](#) (2010)
 - many more
2. Learn weights associated with log normalized counts and use limma
 - [limma-voom](#) (2014)

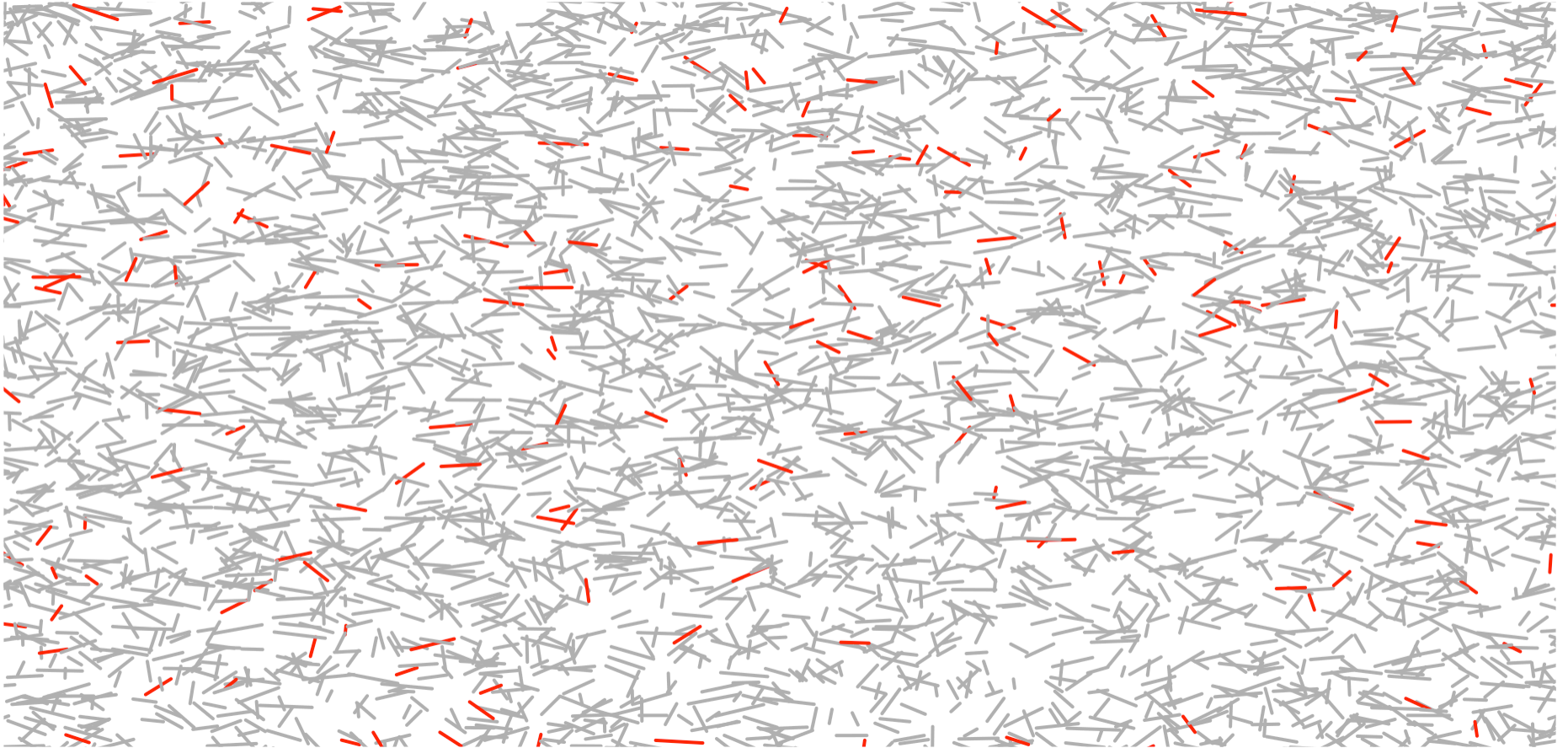
Important for statistical analysis

- Total number of fragments is technical artifact
- Heteroskedasticity of counts
- Each gene has different variability
- Technical covariates can not be excluded!
- With $n > 20$, estimate and include technical factors in the design

Total number of fragments

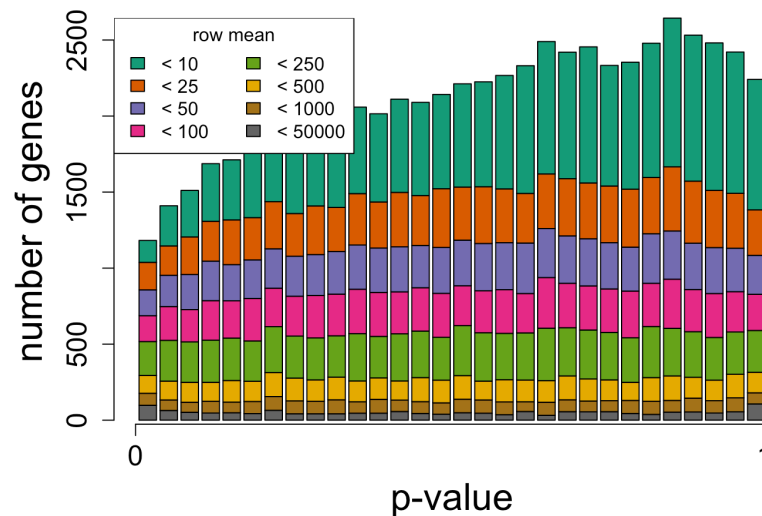


Sampling fragments



Poisson across technical replicates

- From [Bullard 2010](#) take 7 technical replicates
- Calculate expected value $\hat{\lambda}_{ij}$ using DESeq2 norm
- $P(K_{ij} < \hat{\lambda}_{ij})$ assuming $K_{ij} \sim \text{Pois}(\hat{\lambda}_{ij})$



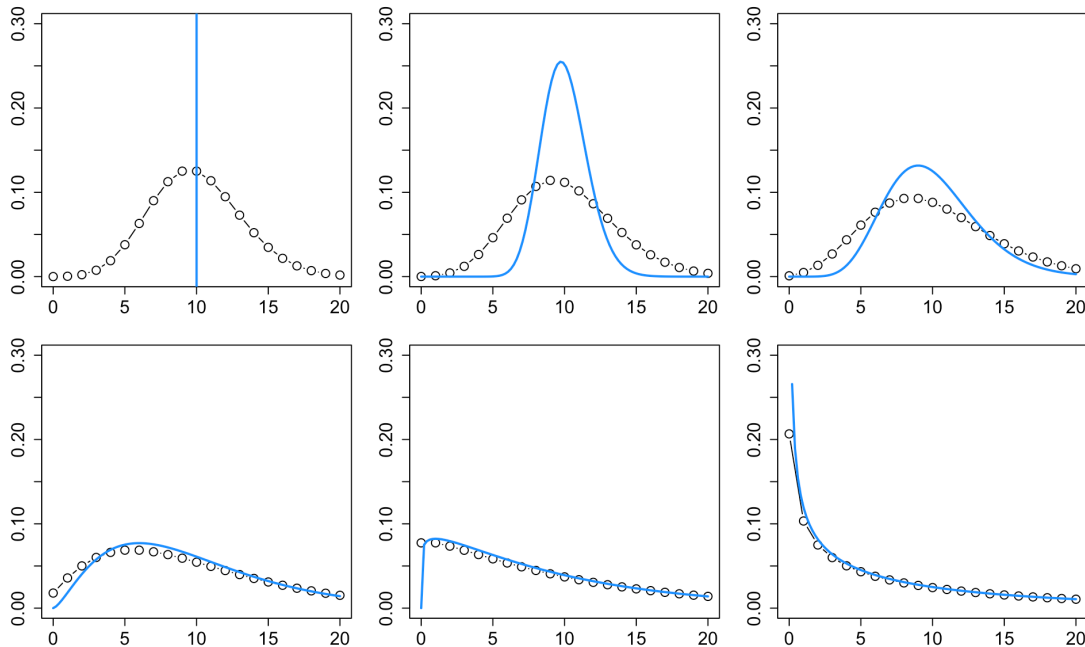
**However, expression not equal across
biological replicates**



Negative Binomial / Gamma Poisson

$$K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$$

$$\text{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$



NB model for RNA-seq

- s_j deals with sequencing depth

$$K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$$

$$\mu_{ij} = s_j q_{ij}$$

$$\log_2(q_{ij}) = \beta_{i0}, \quad j \in A$$

$$\log_2(q_{ij}) = \beta_{i0} + \delta_i, \quad j \in B$$

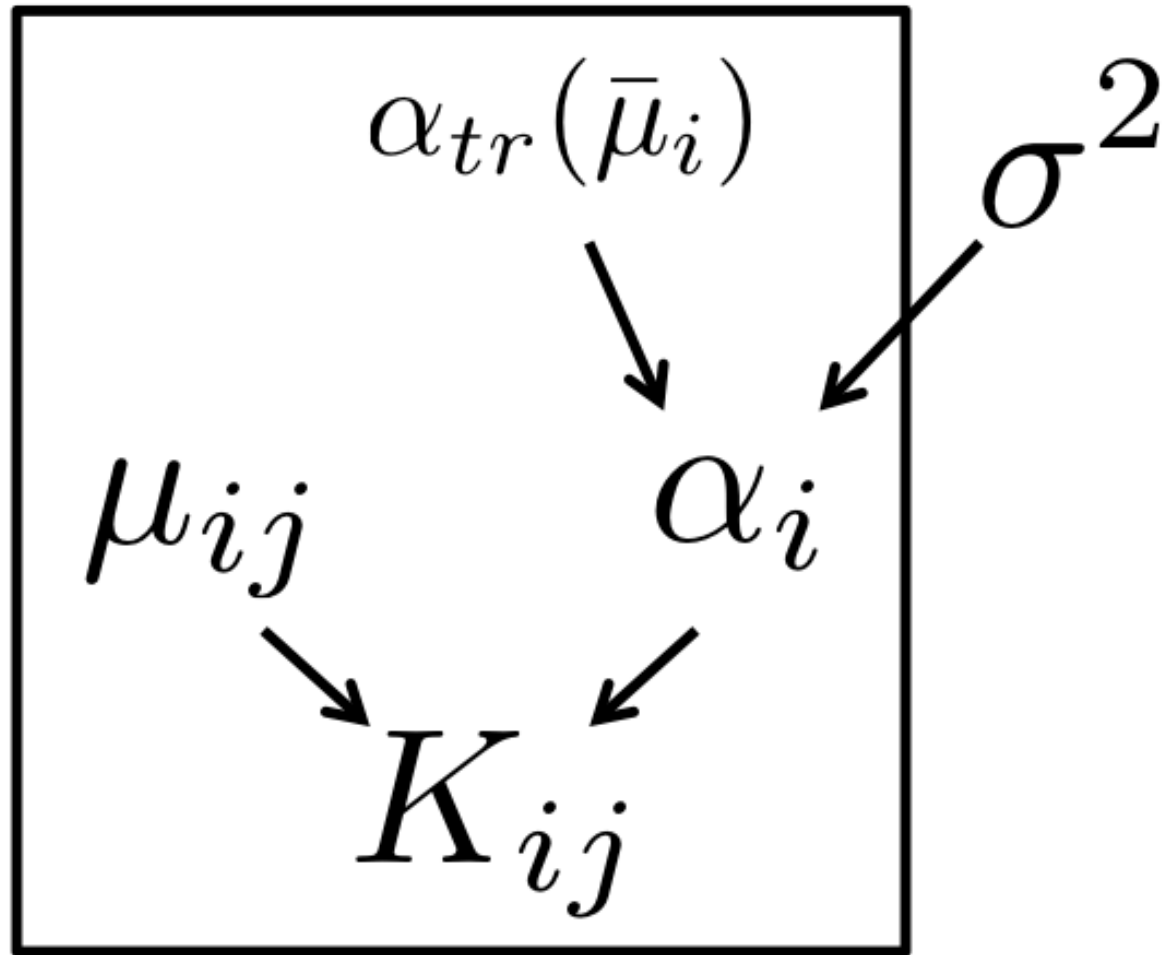
$\delta_i \neq 0$ implies DE (differential expression)

Moderation of dispersion

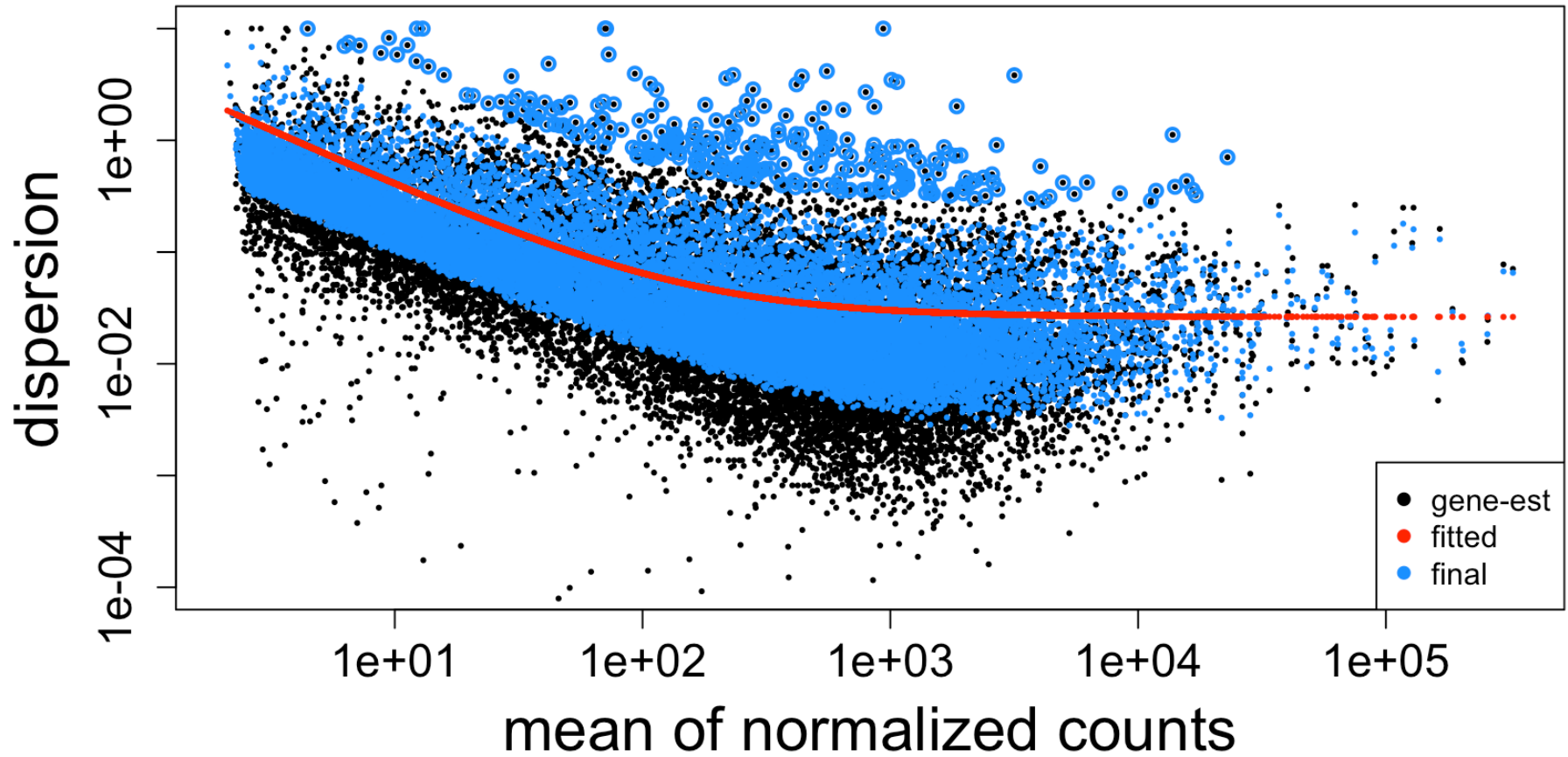
- In [DESeq2](#), a prior on $\log(\alpha_i)$
- Calculate the mean of normalized counts $\bar{\mu}_i$
- A trend line of dispersion over mean $\alpha_{tr}(\mu)$
- Width of the prior σ^2 estimated via assumption of normal sampling variance of $\log(\hat{\alpha}_i)$

$$\log(\alpha_i) \sim N(\log(\alpha_{tr}(\bar{\mu}_i)), \sigma^2)$$

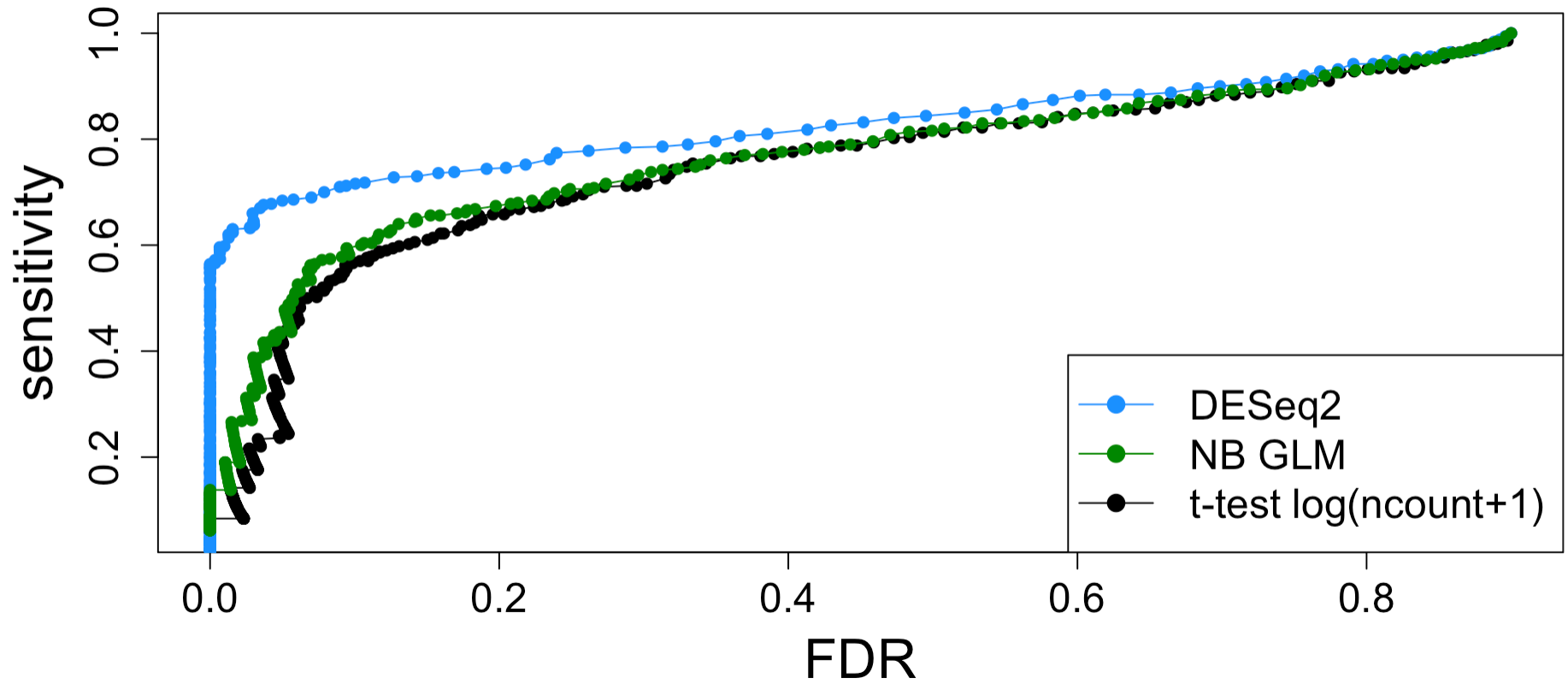
Moderation of dispersion



Moderation of dispersion



Evaluate via simulation



Summary

- Model for counts similar to the hierarchical linear model, but constructed for the dispersion parameter
- Final dispersion estimate plug-in value in DESeq2, edgeR
- [edgeR quasi-likelihood](#) takes into account dispersion estimation uncertainty
- [limma-voom](#) uses weights on log normalized counts
- [apeglm](#) uses Cauchy prior (heavy tails) on effect sizes

Questions?